9.6 Inscribed Triangles & Quadrilaterals

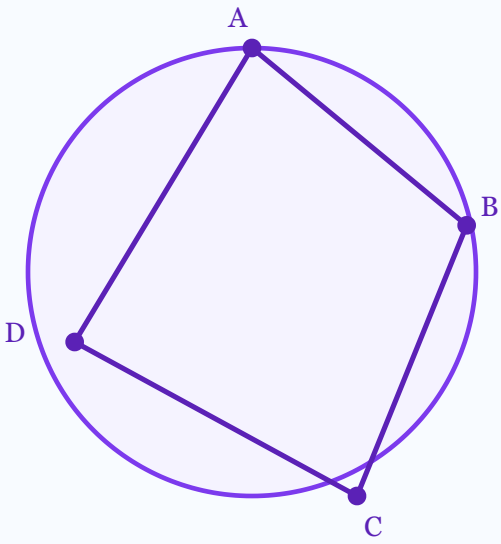
A polygon is **cyclic** when all of its vertices lie on a single circle. For a cyclic quadrilateral, opposite angles are always **supplementary** ($180°$ total) — this follows directly from the Inscribed Angle Theorem, since each pair of opposite angles intercepts a pair of arcs that together make up the entire $360°$ circle. The converse is also true and useful: if a quadrilateral's opposite angles are supplementary, it can be inscribed in a circle. **Thales' Theorem** (from the previous lesson) is the triangle version of this idea: an inscribed triangle with one side as a diameter always has a right angle opposite that diameter. This worksheet combines both ideas, including the proof of the cyclic-quadrilateral rule and problems that mix arcs, angles, and algebra.

Examples

Example 1: Opposite angles are supplementary

Cyclic quadrilateral $ABCD$ has $m\angle A = 85°$. Find $m\angle C$.

Solution



Opposite angles A and C intercept arcs that together fill the whole circle.

Opposite angles of a cyclic quadrilateral sum to $180°$:

$m\angle C = 180° - 85° = 95°$.

Answer: $95°$.

Example 2: Solve for x

Cyclic quadrilateral $ABCD$ has $m\angle B = (3x + 10)°$ and $m\angle D = (2x + 30)°$ (opposite angles). Find x and both angles.

Solution

Identify that $\angle B$ and $\angle D$ are opposite angles of a cyclic quadrilateral, so they are supplementary.

Set up and solve the equation:

$(3x + 10) + (2x + 30) = 180$

$5x + 40 = 180$

$5x = 140 \Rightarrow x = 28.$

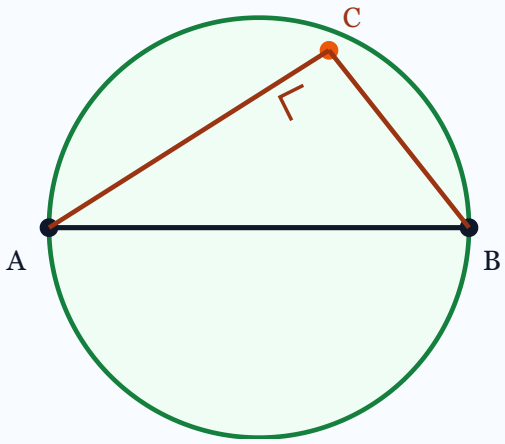
$m\angle B = 3(28) + 10 = 94°$; $m\angle D = 2(28) + 30 = 86°$.

Check: $94 + 86 = 180°$. ✓

Example 3: Thales' Theorem in a cyclic figure

Triangle ABC is inscribed in a circle with diameter $\overline{AC}$. $m\angle A = 40°$. Find $m\angle C$.

Solution



The angle opposite a diameter is always 90°.

$m\angle B = 90°$ (Thales' Theorem), so $m\angle C = 180° - 90° - 40° = 50°$.

Answer: 50°.

Example 4: Arcs and an inscribed angle

Cyclic quadrilateral $ABCD$ has arcs $m\overset{\frown}{AB} = 80°$, $m\overset{\frown}{BC} = 95°$, $m\overset{\frown}{CD} = x$, $m\overset{\frown}{DA} = 110°$. Find x and $m\angle B$.

Solution

All four arcs sum to $360°$: $80 + 95 + x + 110 = 360 \Rightarrow 285 + x = 360 \Rightarrow x = 75°$.

$\angle B$ is inscribed and intercepts arc ADC (the arc from A to C through D, not containing B):

$m\overset{\frown}{ADC} = m\overset{\frown}{AD} + m\overset{\frown}{DC} = 110° + 75° = 185°$.

$m\angle B = \frac{1}{2}(185°) = 92.5°$.

Answer: $x = 75°$; $m\angle B = 92.5°$.

Example 5: Can it be cyclic?

A quadrilateral has $m\angle A = 70°$ and $m\angle C = 100°$ (opposite angles). Can it be inscribed in a circle?

Solution

Check the supplementary requirement: $70° + 100° = 170° \neq 180°$.

Answer: No — the opposite angles don't sum to $180°$, so this quadrilateral cannot be inscribed in a circle.

Example 6: Opposite angles

A cyclic quadrilateral has $m\angle A = 112°$. Find $m\angle C$.

Solution

Identify that $\angle A$ and $\angle C$ are opposite angles of a cyclic quadrilateral, so they must be supplementary.

Subtract from $180°$:

$180° - 112° = 68°$.

Answer: $68°$.

Example 7: Solve for x

$m\angle B = 4x$ and $m\angle D = (2x + 30)^\circ$ (opposite angles). Find x and both angles.

Solution

Identify that opposite angles of a cyclic quadrilateral sum to 180°.

Set up and solve the equation:

$4x + 2x + 30 = 180$

$6x = 150 \Rightarrow x = 25$.

$m\angle B = 4(25) = 100^\circ$; $m\angle D = 2(25) + 30 = 80^\circ$.

Answer: $x = 25$; angles 100° and 80°.

Example 8: Thales — algebra

A right triangle is inscribed with the hypotenuse as a diameter; the other two angles are $3x$ and $2x$. Find x and both angles.

Solution

$3x + 2x = 90^\circ$ (the two acute angles must complete the 90° right angle to reach 180° total):

$5x = 90^\circ \Rightarrow x = 18^\circ$.

Angles: 54° and 36°.

Answer: $x = 18^\circ$; angles 54° and 36°.

Example 9: Cyclic test

A quadrilateral has opposite angles 95° and 85°. Can it be cyclic?

Solution

$95^\circ + 85^\circ = 180^\circ$.

Answer: Yes — the angles are supplementary, so the quadrilateral can be inscribed in a circle.

Example 10: Proving opposite angles are supplementary

Prove that opposite angles $\angle A$ and $\angle C$ of cyclic quadrilateral $ABCD$ are supplementary.

Solution

$\angle A$ is an inscribed angle intercepting arc BCD (the arc from B to D passing through C, which does not contain A): $m\angle A = \frac{1}{2} m\widehat{BCD}$.

$\angle C$ is an inscribed angle intercepting arc BAD (the arc from B to D passing through A, which does not contain C): $m\angle C = \frac{1}{2} m\widehat{BAD}$.

Arcs BCD and BAD together make up the entire circle: $m\widehat{BCD} + m\widehat{BAD} = 360°$.

Adding the two angle equations: $m\angle A + m\angle C = \frac{1}{2}(m\widehat{BCD} + m\widehat{BAD}) = \frac{1}{2}(360°) = 180°$.

The same argument (swapping which pair of arcs you use) proves $\angle B$ and $\angle D$ are supplementary too.

Example 11: Application — a circular skylight frame

A circular skylight is framed by quadrilateral $PQRS$ inscribed in the circle. A carpenter cuts corner P at $118°$. What angle should the opposite corner R be cut at so the frame fits the circular opening?

Solution

Opposite corners of a cyclic quadrilateral must be supplementary:

$m\angle R = 180° - 118° = 62°$.

Answer: $62°$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Opposite angles

Cyclic quadrilateral has $m\angle A = 78°$. Find $m\angle C$.

Answer: **Verification:**

Problem 2: Opposite angles

Cyclic quadrilateral has $m\angle B = 105°$. Find $m\angle D$.

Answer: **Verification:**

Problem 3: Solve for x

$m\angle A = 5x$ and $m\angle C = (3x + 20)°$ (opposite). Find x and both angles.

Answer: **Verification:**

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Problem 4: Thales

Diameter $\overline{AB}$, point C on circle. $m\angle A = 50°$. Find $m\angle B$.

Answer: **Verification:**

Problem 5: Cyclic test

Opposite angles $110°$ and $70°$. Can it be cyclic?

Answer: **Verification:**

Problem 6: Cyclic test

Opposite angles $88°$ and $95°$. Can it be cyclic?

Answer: **Verification:**

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Problem 7: Solve for x

$m\angle B = (2x + 10)°$ and $m\angle D = (4x - 4)°$ (opposite). Find x and both angles.

Answer: **Verification:**

Problem 8: Arc sum

Arcs of a cyclic quad: $70°$, $100°$, $85°$, and x. Find x.

Answer: **Verification:**

Problem 9: Inscribed angle

In a cyclic quad, $\angle A$ intercepts arc $BCD = 160°$. Find $m\angle A$.

Answer: **Verification:**

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Problem 10: Thales

Inscribed right triangle; both acute angles equal. Find each acute angle.

Answer: **Verification:**

Problem 11: Opposite angles

$m\angle A = 125°$. Find $m\angle C$.

Answer: **Verification:**

Problem 12: Vocabulary

What must be true of opposite angles in a cyclic quadrilateral?

Answer: **Verification:**

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Problem 13: Thales

Diameter hypotenuse; one acute angle is $32°$. Find the other.

Answer: **Verification:**

Problem 14: Challenge

Cyclic quadrilateral: $m\angle A = (3x + 20)°$ and $m\angle C = (2x + 10)°$ (opposite). Find x, $m\angle A$, and $m\angle C$. Then, if $m\angle B = (4x - 15)°$, find $m\angle D$.

Answer: **Verification:**

Problem 15: Challenge

In cyclic quadrilateral $ABCD$, diagonal $\overline{AC}$ is a diameter of the circle. Explain why both $m\angle ABC$ and $m\angle ADC$ must equal $90°$.

Answer: **Verification:**

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✓ Answer Key

1. 102°
2. 75°
3. $x = 20$; angles 100°, 80°
4. 40°
5. Yes
6. No
7. $x = 29$; angles 68°, 112°
8. 105°
9. 80°
10. 45° each
11. 55°
12. They are supplementary (sum to 180°)
13. 58°
14. $x = 30$; $m\angle A = 110°$, $m\angle C = 70°$; $m\angle B = 105°$, so $m\angle D = 75°$
15. $\angle ABC = 90°$ by Thales' Theorem; since $ABCD$ is cyclic, $\angle ADC = 180° - 90° = 90°$ too

Great work — keep practising!